

Supervised Learning with Quantum-Enhanced Feature Spaces

Authors: Vojtech Havlicek, Antonio D. Corcoles, Jay M. Gambetta

Published in: Nature 567, 209-212 (2019) | Archival ID: QV-QC-41

1. ABSTRACT & THEORETICAL FOUNDATION

We implement quantum support vector machines and kernel methods on superconducting hardware, mapping classical vectors non-linearly into exponentially large Hilbert spaces for provable quantum classification.

2. CORE CONCEPTUAL INTUITION (PLAIN ENGLISH)

"If you have red and blue marbles tangled together on a flat table, you can't draw a straight line between them. But if you throw them into the air (a higher dimension), you can slice a sheet of paper between them."

3. Mathematical Formulation & Unitary Rule

$$K(\vec{x}_i, \vec{x}_j) = |\langle \theta^{\otimes n} | U_{\Phi}^{\dagger}(\vec{x}_i) U_{\Phi}(\vec{x}_j) | \theta^{\otimes n} \rangle|$$

The quantum kernel computes transition probability between state vectors in 2^n dimensional Hilbert space.

4. Step-by-Step Circuit Sequence

ZZFeatureMap circuit encoding classical features into single- and two-qubit Pauli-Z rotations.

5. Executable IBM Qiskit (Python) Code

```
from qiskit.circuit.library import ZZFeatureMap
from qiskit_machine_learning.kernels import FidelityQuantumKernel

# Encode 2-dimensional classical data into 2 qubits
feature_map = ZZFeatureMap(feature_dimension=2, reps=2, entanglement='linear')
quantum_kernel = FidelityQuantumKernel(feature_map=feature_map)
print("Quantum kernel initialized with ZZ feature map.")
```

6. Computational Complexity & Speedup

Classical Time:

Quantum Time:

Speedup Class:

Exponential time $O(2^n)$ to compute $O(\text{poly}(n))$ of classical depth and $O(M^2/\epsilon)$ exponential speedup at sampling shots for M data points.

7. Key Applications & Future Research Directions

- Credit fraud and high-frequency financial risk classification
- High-energy particle collision event classification at CERN LHC
- Genomics sequence analysis and cancer subtype biomarker detection

Future Work:

Kernel matrix approximation, quantum data embedding design, and barren plateau avoidance in quantum neural networks.